

PROJECT DOCUMENTATION

MAVYN

Data Cleaning, Modeling & Business Intelligence

Team Members

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Tools Used: Power BI | Microsoft Excel | Tableau

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1. Project Planning & Management

1.1 Project Proposal

This project focuses on analyzing MAVYN's retail sales data to extract meaningful business insights using Power BI, Microsoft Excel, and Tableau. The dataset encompasses transactional data from 35 stores across multiple countries, including sales, returns, discounts, customer information, and product details.

Objectives

- Clean and prepare raw data across five source tables: transactions, products, stores, customers, and discounts.
- Build a well-structured star-schema data model with proper relationships between tables.
- Create DAX measures and calculated columns to support analytical reporting.
- Develop interactive dashboards in Power BI and Tableau to visualize key KPIs.
- Perform supplementary analysis using Microsoft Excel.

Scope

- Data sources: 5 CSV files (transactions, products, stores, customers, discounts).
- Geographic coverage: United States, China, Germany, United Kingdom, France, Spain, Portugal.
- Time period: 2020 – 2024.
- Total records: 6,416,827 transactions, 783,217 customers, 17,940 products, 35 stores, 181 discount records.

1.2 Project Plan

Phase	Task	Responsible	Duration
Phase 1	Data Cleaning & Quality Assessment	Zyad Mohamed	Week 1
Phase 1	Data Modeling & Relationships	Zyad Mohamed	Week 1
Phase 2	Measures, Calculated Columns & DAX	Sahar & Joyce	Week 2
Phase 2	Excel Analysis & Reports	Mahmoud	Week 2
Phase 3	Power BI Dashboard Development	All Team	Week 3
Phase 3	Tableau Dashboard Development	Fathy	Week 3
Phase 4	Testing, QA & Documentation	All Team	Week 4

1.3 Task Assignment & Roles

Team Member	Role	Responsibilities
Zyad Mohamed Alaaeldien	Data Engineer	Data Cleaning, Power Query, Data Modeling, Power BI Dashboards
Sahar Alaa Mohamed	BI Developer	Measures, Calculated Columns, DAX, Power BI Dashboards
Mahmoud Osama Khattab	Data Analyst	Excel Analysis, Reports, Power BI Dashboards
Joyce Nardeen Nabil	BI Developer	Measures, Calculated Columns, DAX, Power BI Dashboards
Fathy Raed Muhammad	BI Developer	Tableau Dashboards, Power BI Dashboards

All team members collaborated on the Power BI dashboard development.

1.4 Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
Encoding issues in CSV files	High	High	Apply UTF-8 encoding fix in Power Query Source step
Many-to-many relationships	Medium	High	Use DAX-based date range logic instead of direct joins
Large file size (700MB+)	High	Medium	Work with distinct-value lookups; transform in Power Query
Ambiguous relationship paths	Medium	High	Set correct Active/Inactive relationship priorities
Missing or NULL values	High	Medium	Analyze context before replacing — keep intentional NULLs
Multi-currency figures	High	High	Standardize all monetary columns to USD via fixed exchange rate DAX columns

1.5 KPIs (Key Performance Indicators)

- Total Revenue (USD, net of returns)
- Return / Refund Rate by Product Category and Store
- Discount Impact on Revenue and Gross Margin
- Customer Lifetime Value (CLV) and Purchase Frequency
- New vs Returning Customer split
- Top Selling Products and Categories
- Customer Demographics — Age, Generation, Job Category
- Store & Employee Performance

2. Literature Review

The field of retail business intelligence has seen significant growth with the adoption of self-service BI tools such as Power BI, Tableau, and Excel. The following references informed the methodological approach of this project:

- Microsoft Power BI Documentation — Data Modeling Best Practices (docs.microsoft.com)
- Tableau Public Knowledge Base — Dashboard Design Principles (help.tableau.com)
- Kimball, R. & Ross, M. (2013). The Data Warehouse Toolkit, 3rd Edition — Star Schema design principles applied to the data model.
- DAMA International (2017). DAMA-DMBOK: Data Management Body of Knowledge — Data quality assessment framework.
- McKinsey Global Institute — The Age of Analytics: Competing in a Data-Driven World (2016).

Key concepts applied from the literature include: star schema modeling, data profiling techniques, handling of multi-currency datasets, customer lifetime value (CLV) modeling, and best practices for dealing with NULL values in analytical contexts.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Interest	Key Needs
Business Management	High	Revenue trends, store performance, profitability
Sales Team	Medium	Product performance, discount effectiveness
Operations	Medium	Store staffing, return rates, inventory insights
Marketing Team	Medium	Customer segmentation (Generation, Job Category, CLV)
Data Team	High	Clean data, reliable model, reusable measures
Academic Reviewer	High	Complete documentation, methodology, reproducibility

3.2 Functional Requirements

- The system shall load and clean data from 5 CSV source files.
- The system shall handle multi-language and multi-currency data correctly.
- The system shall establish correct relationships between all tables.
- The system shall support filtering by date, country, store, category, and product.
- The system shall calculate revenue net of returns using DAX measures, standardized to USD.
- The system shall apply discounts based on date range logic.
- The system shall segment customers by age, generation, job category, and purchase behavior.
- The system shall provide interactive dashboards in Power BI and Tableau.

3.3 Non-Functional Requirements

- Performance: Dashboards shall load within acceptable time despite 6.4M+ transaction rows.
- Usability: Dashboards shall be intuitive for non-technical business users.
- Reliability: Data model shall produce consistent, accurate results across tools.
- Maintainability: Power Query steps and DAX measures shall be clearly named and documented.
- Scalability: Model design shall support additional years of data without restructuring.

4. System Analysis & Design

4.1 Problem Statement & Objectives

MAVYN operates a global fashion retail chain with 35 stores across 7 countries. Raw transactional data was collected in CSV format from multiple regions, resulting in data quality issues including encoding errors, missing values, non-English names, and inconsistent currency symbols. The objective is to transform this raw data into a clean, reliable analytical model that supports strategic business decision-making.

4.2 Data Sources Overview

Table	Rows	Description
transactions	6,416,827	Sales and return transactions with line/invoice totals, currency, discount
customers	783,217	Customer demographics: name, city, country, DOB, job title, gender
products	17,940	Product catalog with descriptions, sizes, colors, production cost
stores	35	Store locations with city, country, ZIP code, employee count
discounts	181	Storewide promotional discounts with date ranges and descriptions
employees	~300	Employee records linked to stores

4.3 Data Quality Issues & Resolutions (Summary)

Table	Issue	Resolution
transactions	Currency symbol encoding ('¥', '£', '€', 'â', 'ä', 'ä', 'ä')	Replace Values → '¥', '£', '€' in Power Query
transactions	Negative Line/Invoice Total	Kept — confirmed Return transactions, required for net revenue
products	NULL Color	Kept — product sold in a single fixed color, not a variant
products	NULL Sizes	Replaced with 'One Size' — confirmed 100% Accessories category
stores	Non-English Country/City/Store Name	Translated to English via AI translation tool (extracted Store Name/City → translated → imported → merged into stores table)
employees	Non-English Employee Names	Extracted → translated via AI translation tool → imported → added into the employees table
discounts	NULL Category/Sub Category	Kept — intentional storewide promotions (Black Friday, etc.)
customers	Encoding broken (Name, City, Country)	File Origin set to 65001 (UTF-8) in Power Query Source step

Table	Issue	Resolution
customers	Chinese names unreadable for reporting	Transliterated to Pinyin via Python pypinyin (Name_EN column)
customers	Chinese city names	Translated to English (City_EN column) via mapping table
customers	Blank Country for Spanish cities	Filled with 'Spain' after confirming all blank-country cities were Spanish

4.4 Data Model & Relationships

The data model follows a Star Schema design with the Transactions table as the central Fact Table, surrounded by Dimension Tables.

From Table (Column)	Relationship	To Table (Column)	Status
transactions (Customer ID)	Many → 1	customers (Customer ID)	Active
transactions (Product ID)	Many → 1	products (Product ID)	Active
transactions (Employee ID)	Many → 1	employees (Employee ID)	Active
transactions (Store ID)	Many → 1	stores (Store ID)	Active
employees (Store ID)	Many → 1	stores (Store ID)	Inactive
discounts	—	transactions	No direct join — handled via DAX date logic

Key Modeling Decisions

- transactions → stores set as Active (primary relationship for store analysis).
- employees → stores set as Inactive to resolve ambiguous path through employees table.
- Discounts table kept disconnected from the model and evaluated via DAX FILTER on date range.

4.5 Technology Stack

Tool	Purpose	Usage in Project
Power BI Desktop	Primary BI Platform	Data cleaning, modeling, DAX, main dashboard
Power Query (M Language)	ETL / Data Transformation	Cleaning steps, encoding fixes, Replace Values, Merge
DAX	Analytics Language	Calculated columns, measures, KPI logic
Python (pypinyin, openpyxl)	Supporting Data Prep	Chinese name transliteration, bulk translation lookups
Microsoft Excel	Supplementary Analysis	Pivot tables, translation lookups, supporting charts
Tableau	Secondary Visualization	Alternative dashboard with advanced visual analytics

5. Implementation

5.1 Power Query — Data Cleaning Steps (Summary)

- transactions: Source → Promoted Headers → Changed Type → Replaced Value (¥) → Replaced Value1 (£) → Replaced Value2 (€).
- products: Source → Promoted Headers → Changed Type → Replaced Value (Sizes NULL → 'One Size').
- stores: Source → Promoted Headers → Changed Type → Extracted Store Name and City columns (multilingual/inconsistent values) → Sent to an AI translation tool → Imported translated values → Added translated columns into the stores table (replacing the previous Replace Values approach).
- customers: Source (File Origin 65001 UTF-8) → Promoted Headers → Changed Type → Merge with translation lookup tables (Name_EN, City_EN, Country) → Remove Duplicates on lookup tables before merging.
- employees: Source → Promoted Headers → Changed Type → Extracted Employee Name column → Sent to an AI translation tool → Imported translated names → Added the translated column into the employees table.

5.2 DAX Calculated Columns

Transactions Table

Type of transactions

```
Type of transactions =  
IF(Transactions[Discount] = 0, "Full price",  
  IF(Transactions[Discount] > 0, "Discounted"))
```

Invoice Total USD — converts Invoice Total to USD using fixed exchange rates

```
Invoice Total USD =  
SWITCH(  
    transactions[Currency Symbol],  
    "$", transactions[Invoice Total] * 1,  
    "€", transactions[Invoice Total] * 1.0939,  
    "£", transactions[Invoice Total] * 1.30,  
    "¥", transactions[Invoice Total] * 0.1383,  
    transactions[Invoice Total] * 0.14  
)
```

Line Total USD — same logic applied to Line Total

```
Line Total USD =  
SWITCH(  
    transactions[Currency Symbol],  
    "$", transactions[Line Total] * 1,  
    "€", transactions[Line Total] * 1.0939,  
    "£", transactions[Line Total] * 1.30,  
    "¥", transactions[Line Total] * 0.1383,  
    transactions[Line Total] * 0.14  
)
```

Day Type — classifies each transaction date as Weekday or Weekend

```
Day Type =  
IF(  
    WEEKDAY('transactions'[Date], 2) > 5,  
    "Weekend",  
    "Weekday"  
)
```

Customer Type — flags each transaction as the customer's first purchase (New) or a later one (Returning)

```
Customer Type =  
VAR FirstPurchase =  
    CALCULATE(  
        MIN(transactions[Date]),  
        ALLEXCEPT(transactions, transactions[Customer ID])  
    )  
RETURN  
    IF(transactions[Date] = FirstPurchase, "New", "Returning")
```

Discount Bracket — groups discount percentage into readable ranges

```
Discount Bracket =  
SWITCH(  
    TRUE(),  
    Transactions[Discount] = 0, "0%",  
    Transactions[Discount] > 0 && Transactions[Discount] <= 0.20, "10%-20%",  
    Transactions[Discount] > 0.20 && Transactions[Discount] <= 0.30, "20%-30%",  
    Transactions[Discount] > 0.30 && Transactions[Discount] <= 0.40, "30%-40%",  
    Transactions[Discount] > 0.40, "40%+"  
)
```

Customers Table

Job Category — groups Job Title into 17 broader job categories via SWITCH/IN logic (sample: Agriculture & Environment)

```
Job Category =  
SWITCH(  
    TRUE(),  
    -- Agriculture & Environment  
    [Job Title] IN {  
        "Agricultural consultant", "Agricultural engineer",  
        "Animal nutritionist", "Animal technologist",  
        "Ecologist", "Environmental consultant",  
        "Environmental education officer",  
        "Environmental health practitioner", "Farm manager",  
        "Fish farm manager", "Fisheries officer",  
        "Nature conservation officer", "Operations geologist",  
        "Rural practice surveyor", "Soil scientist",  
        "Water quality scientist", "Wellsite geologist",  
        "Geologist, engineering" ...  
    } ...  
    -- (16 more categories follow: Armed Forces & Security,  
    -- Arts Media & Design, Construction & Surveying, Education,  
    -- Engineering, Finance & Accounting, Healthcare & Medicine,  
    -- Hospitality & Food, IT & Technology, Legal & Law,  
    -- Management & Administration, Sales & Marketing,  
    -- Science & Research, Social & Community Services,  
    -- Students, Other)  
)
```

Customer Lifespan (Month) — months between a customer's first and last purchase

```
Customer Lifespan (Month) =  
VAR FirstPurchase =  
    CALCULATE(  
        MIN(transactions[Date]),  
        ALLEXCEPT(transactions, transactions[Customer ID])  
    )  
VAR LastPurchase =  
    CALCULATE(  
        MAX(transactions[Date]),  
        ALLEXCEPT(transactions, transactions[Customer ID])  
    )  
RETURN  
    DATEDIFF(FirstPurchase, LastPurchase, MONTH)
```

Age — customer's current age based on Date Of Birth

```
Age = DATEDIFF(customers[Date Of Birth], TODAY(), YEAR)
```

Generation — classifies customers into generational cohorts by Age

```
Generation =  
SWITCH(  
    TRUE(),  
    customers[Age] >= 60, "Boomer+",  
    customers[Age] >= 44, "Gen X",  
    customers[Age] >= 28, "Millennial",  
    "Gen Z"  
)
```

Transactions Count — number of transactions associated with each customer

```
Transactions Count = COUNTROWS(RELATEDTABLE(transactions))
```

Discounts Table

Sale Name — extracts the promotion name from the Description text

```
Sale Name =  
TRIM(  
    MID(  
        [Description],  
        FIND(" our ", [Description]) + 5,  
        LEN([Description]) - FIND(" our ", [Description]) - 4  
    )  
)
```

Days of Sale — length of each promotional period in days

```
Days of Sale = DATEDIFF([Start], [End], DAY) + 1
```

5.3 DAX Measures

All measures are organized in a dedicated 'measure' table for clarity.

Revenue & Profitability

Measure	DAX Formula
Total Revenue	SUM(transactions[line Total USD])
Total Return Value	ABS(CALCULATE([Total Revenue], transactions[Transaction Type]="return"))
Total Revenue under Discount	CALCULATE(SUM(transactions[line Total USD]), transactions[Type of transactions]="Discounted")
Total Discounted Amount	(SUMX(transactions, transactions[Quantity] * transactions[Unit Price]) - [Total Revenue])
COGS	SUMX(transactions, RELATED(products[Production Cost]) * transactions[Quantity])
Total profit	[Total Revenue] - [COGS]
Gross Margin %	DIVIDE([Total profit], [Total Revenue], 0)
P_M of Discounted Sales	CALCULATE([Gross Margin %], transactions[Type of transactions]="Discounted")
P_M of Full Price sales	CALCULATE([Gross Margin %], transactions[Type of transactions]="Full price")
AVG profit per invoice	[Total profit] / [Total transactions]

Transactions & Order Metrics

Measure	DAX Formula
Total transactions	DISTINCTCOUNT(transactions[Invoice ID])
Total quantity	SUM(transactions[Quantity])
AOV (Average Order Value)	DIVIDE([Total Revenue], [Total transactions], 0)
AVG unit per invoice	DIVIDE(SUM(transactions[Quantity]), [Total transactions], 0)
AVG Given Discount%	CALCULATE(AVERAGE(transactions[Discount]), transactions[Discount]<>0.00)
Count of discounts	CALCULATE(COUNT(transactions[Discount]), transactions[Discount]<>0.00)
Refund count	CALCULATE(DISTINCTCOUNT(transactions[Invoice ID]), transactions[Transaction Type]="Return")
Refund rate %	DIVIDE([Refund count], [Total transactions])

Customer Metrics

Measure	DAX Formula
Active customers	DISTINCTCOUNT(transactions[Customer ID])
Total customers	(DISTINCTCOUNT(customers[Customer ID])) + [New Customers]
New Customers	COUNTROWS(EXCEPT(DISTINCT(transactions[Customer ID]), DISTINCT(customers[Customer ID])))
Purchase Frequency	DIVIDE([Total transactions], [Active customers])
Repeat Purchase Rate %	Customers with >1 invoice ÷ Total distinct customers (via FILTER + VALUES + CALCULATE)
CLV (Customer Lifetime Value)	[AOV] * [Purchase Frequency] * [Gross Margin %] * [AVG CL(month)]

Store, Employee & Catalog Metrics

Measure	DAX Formula
Countries	DISTINCTCOUNT(stores[Country])
employees	DISTINCTCOUNT(employees[Employee ID])
Active Employees	DISTINCTCOUNT(transactions[Employee ID])
Total Products in Catalog	DISTINCTCOUNT(products[Product ID])

New Customers — Full Formula

```

New Customers =
VAR TransactionCustomers =
    DISTINCT(transactions[Customer ID])
VAR RegisteredCustomers =
    DISTINCT(customers[Customer ID])
RETURN
    COUNTROWS(EXCEPT(TransactionCustomers, RegisteredCustomers))

```

Repeat Purchase Rate % — Full Formula

```

Repeat Purchase Rate % =
VAR CustomersWithRepeat =
    CALCULATE(
        DISTINCTCOUNT(transactions[Customer ID]),
        FILTER(
            VALUES(transactions[Customer ID]),
            CALCULATE(DISTINCTCOUNT(transactions[Invoice ID])) > 1
        )
    )
VAR TotalCustomers =
    DISTINCTCOUNT(transactions[Customer ID])
RETURN
    DIVIDE(CustomersWithRepeat, TotalCustomers)

```

6. Testing & Quality Assurance

6.1 Data Quality Checks

Check	Method	Result
NULL values per column	Column profiling in Power Query	All intentional NULLs documented and justified
Currency symbol encoding	Filter Currency Symbol column	3 encoding errors identified and fixed
Negative transaction values	Sort Line Total ascending	Confirmed Return transactions — valid
Duplicate Product IDs	Group By Product ID, count rows	Each product has exactly 1 row — no duplicates
Store name consistency	Filter Store Name column	All non-English names translated to English
Relationship integrity	Manage Relationships dialog	All cardinalities verified as Many-to-1
Currency conversion accuracy	Spot-check Invoice Total USD vs Currency Symbol	Correct rate applied per currency
Customer encoding fix	Sample Chinese/German names post-fix	Readable characters, transliteration verified

6.2 Model Validation

- All Active relationships verified as Many-to-1 with single direction filter.
- discounts → transactions relationship removed and replaced with DAX FILTER logic.
- employees → stores set as Inactive to resolve ambiguous path.
- transactions → stores confirmed as Active primary relationship.

6.3 Measure Validation

- Total Revenue cross-checked against SUM of raw Line Total converted manually for a sample of currencies.
- Refund rate % and Repeat Purchase Rate % validated against manually filtered subsets in Excel.
- CLV formula components (AOV, Purchase Frequency, Gross Margin %) validated independently before combining.

6.4 Dashboard Testing

- All slicers and filters tested across both Power BI and Tableau dashboards.
- KPI values cross-validated between Power BI measures and Excel calculations.
- Date range filtering verified to correctly apply discount logic.

7. Power BI Dashboards

The Power BI report consists of 11 interactive dashboard pages built on top of the cleaned data model, calculated columns, and DAX measures described in Sections 4–5. All team members collaborated on dashboard development.

- The 'measure' table in Power BI Desktop includes a Count of discounts DAX measure alongside the other measures documented in Section 5.3.
- The customers table includes the translated Name_EN and City_EN columns alongside Age, Generation, Job Category, Customer Lifespan (Month), and Transactions Count, as described in Section 4.3.

7.1 Overview

High-level snapshot of overall business performance.

- KPI cards: Total Revenue (283.46M), Total Profit (161.09M), Gross Margin % (56.83%), Active Customers (1.28M).
- Line chart: Total profit by Year and Quarter (Jan 2023 – Jan 2025).
- Donut chart: Total transactions by Transaction Type (Sale 95.5%, Return 4.47%).
- Table: Category-level breakdown (Feminine, Masculine, Children) showing Total Revenue, Total Quantity, and Total Profit.
- Slicer: Year (All years by default).

7.2 Revenue & Sales Performance

- KPI cards: Total Quantity (7M), Total Transactions (5M), AOV (62.43), AVG Unit per Invoice (1.55).
- Horizontal bar chart: Total Revenue by Store Name (top store: Store New York at 21M).
- Area chart: Total Revenue and Total Profit by Year (2023–2025).
- Column chart: Total Revenue by Generation (Gen Z leads at 60M).

7.3 Transaction Analysis

- KPI cards: Total Revenue (283.46M), Total Transactions (5M), Average of Line Total (114.19).
- Line chart: Total transactions by Year and Month, showing seasonal spikes (e.g. 517K in Oct 2024).
- Horizontal bar chart: Total transactions by Country (United States 1.19M, China 1.10M leading).
- Donut chart: Total transactions by Payment Method (Credit Card 80.03%, Cash 19.97%).

7.4 Returns

- KPI cards: Refund Count (203K), Refund Rate % (4.47%), Total Return Value (16.82M).
- Horizontal bar chart: Refund rate % by City_EN (highest: Weinan at 33.33%).
- Treemap: Total Return Value by Category (Feminine 8.34M, Masculine 7.00M).
- Column chart: Refund count by Size (M highest at 103K).
- Slicer: Year (2023 / 2024 / 2025).

7.5 Stores Performance

- KPI cards: Count of Country (7), Count of City (35), No. of Stores (35).
- Horizontal bar chart: Top Stores by Customers (Store New York leads at 103K).
- Map visual: Total Revenue Breakdown by geographic location (North America, Europe, Asia).
- Combo chart: Top 5 Stores Performance — Total Revenue (columns) vs Active Employees (line).

7.6 Employees

- KPI cards: No. of Employees (404), Positions (5), Active Employees (264).
- Horizontal bar chart: Total Revenue by Employee Name (top performer: 4.48M).
- Combo chart: Total transactions and New Customers by Name.
- Map visual: Number of Employees by Country.

7.7 Customers

- KPI cards: Total Customers (1.47M), New Customers (689K), CLV (3.26K).
- Horizontal bar chart: Top 10 Cities by Customers (Shenzhen leads at 301K).
- Bar chart: Total customers by Job Category (Students 299.17K, Other 173.17K leading).
- Donut chart: Customers by Gender (M 59.11%, F 40.89%).

7.8 Customer Loyalty Analysis

- KPI cards: AVG CL (month) (26.00), Purchase Frequency (3.54), Repeat Purchase Rate % (75.08%).
- Line chart: Total Revenue Over Years by Customer Type (New vs Returning).
- Column chart: Existing Customers' Transactions Count, showing purchase repetition distribution.
- Donut chart: Total Revenue by Customer Type (Returning 69%, New 31%).

7.9 Product Category Performance

- KPI cards: Total Products in Catalog (18K), Total Quantity (7M).
- Horizontal bar chart: Total quantity by Sub Category (Pants and Jeans leads at 687K).
- Donut chart: Total Revenue by Category (Feminine 53.55%, Masculine 46.45%).
- Column chart: Total transactions by Size (M leads at 2.2M).

7.10 Discount Overview

- KPI cards: AVG Given Discount% (42.71%), Total Revenue Under Discount (56.70M), P_M of Discounted Sales (42.04%), P_M of Full Price Sales (60.53%).
- Donut chart: Total transactions by Type of transactions (Full price 71.25%, Discounted 28.75%).
- Donut chart: Discount by Day Type (Weekday 56.4%, Weekend 43.6%).
- Horizontal bar chart: Days of Sale by Sale Name (Mid-Spring Refresh Sale longest at 40 days).
- Column chart: Total transactions by Discount Bracket (0% bracket dominant at 3.3M).

7.11 Profitability & Margin Analysis

- KPI cards: COGS (122.37M), Gross Margin % (56.83%), AVG profit per invoice (35.48).
- Combo chart: Total profit and COGS by Top 10 Stores.
- Column chart: Gross Margin % by Top 3 Countries (China 60.24% leading).
- Waterfall chart: Gross Margin % by Category (Masculine, Feminine, Children contributions).

8. Excel Analysis

Following the Calculated Columns and Measures phase in Power BI, five core reference tables were exported to Excel to support the Stores Overview Dashboard and enable flexible offline analysis.

8.1 Exported Tables

1. Stores Table

Used to evaluate branch performance and investment classification across countries.

- Store ID, Store Name, Country, City, Year
- Store Tier — profit-based classification (Tier A, Tier B, Tier C)
- Quantity, Total Sales USD, Total Profit, Return Rate Percentage, Profit Margin

2. Products Table

Used to analyze product profitability and compare sales against production costs.

- Category, Sub Category
- Quantity, Total Sales USD, Total Production Cost, Total Profit, Return Rate Percentage

3. Product Discounts Table (Product DIS)

Used to monitor the impact of promotional policies on category sales and profit.

- Category, Sub Category
- Sum of Total Sales Before DIS, Sum of Total Sales USD
- Sum of Total Discount Amount, AVG Discount%

4. Employees Table

Used to evaluate sales team performance and identify top performers.

- Employee ID, Name, Store Name, Position
- Quantity, Total Sales USD

5. Forecasting Table

A time-series table used for financial forecasting and daily sales trend analysis.

- Year, Quarter, Month, Day
- Sum of Total Sales USD

8.2 Calculated Columns (Excel Data Model)

Total Sales USD — standardizes Total Sales across all currencies into USD

```
Total Sales USD =  
SWITCH(  
    transactions[Currency Symbol],  
    "$", transactions[Total Sales] * 1,  
    "€", transactions[Total Sales] * 1.08,  
    "£", transactions[Total Sales] * 1.27,  
    "¥", transactions[Total Sales] * 0.0067,  
    transactions[Total Sales] * 0.14  
)
```

Total Profit — net profit after subtracting production cost × quantity from USD sales

```
Total Profit =  
transactions[Total Sales USD] -  
    ( RELATED(products[Production Cost]) * transactions[Quantity] )
```

Year — extracted directly from the transaction date for time-based filtering

```
Year = YEAR(transactions[Date])
```

Unit Price USD — standardizes Unit Price across all currencies into USD

```
Unit Price USD =  
SWITCH(  
    transactions[Currency Symbol],  
    "$", transactions[Unit Price] * 1,  
    "€", transactions[Unit Price] * 1.08,  
    "£", transactions[Unit Price] * 1.27,  
    "¥", transactions[Unit Price] * 0.0067,  
    transactions[Unit Price] * 0.14  
)
```

Total Sales Before DIS — sales value before any discount is applied

```
Total Sales Before DIS =  
transactions[Quantity] * transactions[Unit Price USD]
```

Total Discount Amount — total monetary value of discounts granted per transaction

```
Total Discount Amount =  
transactions[Total Sales Before DIS] - transactions[Total Sales USD]
```

8.3 Measures (Excel Data Model)

Return_Rate_Percentage — return rate as an absolute (non-negative) percentage of sales

```
Return_Rate_Percentage =  
VAR TotalSales =  
    CALCULATE(  
        SUM(transactions[Total Sales USD]),  
        transactions[Transaction Type] = "Sale"  
    )  
VAR TotalReturns =  
    CALCULATE(  
        SUM(transactions[Total Sales USD]),  
        transactions[Transaction Type] = "Return"  
    )  
RETURN  
    DIVIDE(ABS(TotalReturns), TotalSales, 0)
```

Store_Tier — classifies stores into profit-based tiers

```
Store_Tier =  
VAR NetProfit = SUM(transactions[Total Profit])  
RETURN  
    IF(  
        ISBLANK(NetProfit),  
        BLANK(),  
        IF(NetProfit >= 3000000, "Tier A (High Profit)",  
        IF(NetProfit >= 2000000, "Tier B (Medium Profit)",  
        "Tier C (Low Profit)"))  
    )
```

AVG Discount% — average discount rate, excluding zero-discount transactions

```
AVG Discount% =  
CALCULATE(  
    AVERAGE(transactions[Discount]),  
    transactions[Discount] <> 0.00  
)
```

8.4 Stores Overview Dashboard

All exported tables and DAX logic above were consolidated into a single interactive dashboard (Stores Overview Dashboard) for management decision-making.

KPI Cards

- Total Quantity: 7,060,071 units
- Total Sales USD: \$283,462,804.53
- Total Profit: \$161,090,000
- Average of Return_Rate_Percentage: 5.54%

Charts & Visuals

Average Return Rate by Country — horizontal bar chart

- Ranks countries from highest to lowest return rate to identify logistics or shipping issues. Spain ranks highest at 5.67%, followed by the United Kingdom and China; Portugal is lowest at 5.43%.

Profitability Analysis by Store Tier — combo chart (columns & line)

- Compares Total Sales (columns) against Net Profit (line) for each Store Tier. Tier C (Low Profit) shows the highest sales volume (over \$109M) despite a margin that warrants review compared to other tiers.

Country Market Share Distribution by Total Profit — horizontal bar chart

- Ranks countries by net profit achieved.

YOY Profit Growth by Country — line chart

- Tracks year-over-year profit growth between 2023 and 2024 per country, highlighting rising and declining markets: strong growth in Spain (+23.2%) and China (+28.27%), versus a decline in Portugal (−10.45%).

Interactive Slicers

- Store_Tier — filters the entire dashboard by store tier (Tier A, Tier B, Tier C).
- Year — controls and compares data across 2023, 2024, and 2025.

9. Tableau Dashboards

As a secondary visualization deliverable, a Tableau workbook (Final.twb) was built covering the same underlying data with three dashboards focused on financial performance, product/inventory analysis, and customer/geographic insight.

9.1 Data Source & Relationships

The workbook connects to the same seven CSV source files (transactions, customers, employees, invoices, products, stores, discounts) using a Live Connection rather than an Extract, so the data is read directly from the files without a separate extract step.

transactions.csv is the central Fact Table, with 11 fields and 6,416,827 rows.

transactions.csv is joined to customers, employees, invoices, products, and stores through Relationships (not Joins), forming a star schema with five separate connections branching from the fact table. Using Relationships instead of traditional Joins avoids row duplication and inflated totals at aggregation time, since each dimension table keeps its own level of detail and is only combined with transactions.csv on demand, based on the fields used in a given sheet. The discounts.csv file is included among the data sources but is not part of the core relationships.

9.2 Key Fields

Core fields used from transactions.csv include Date, Invoice Total USD, Payment Method (Cash or Credit Card), Transaction Type (Sale or Return), and the Customer ID, Invoice ID, Product ID, Store ID, and Employee keys linking to the dimension tables. The Invoice Total values shown in the raw data represent a single line item within an invoice and are aggregated with SUM at the sheet level to calculate total revenue.

9.3 Financial & Sales Overview Dashboard

Gives a quick overview of overall financial and sales performance, built at Automatic size with a Phone layout for mobile viewing.

KPI	Value
Total Revenue	\$211,956,524.31
Total Transactions	4,540,404
Total Quantity	7,060,071
Total Discount	34,826
Total Customers	783,217
Total Stores	35

- **Monthly Sales Trend:** a line chart showing total sales by month across the year, with an interactive Year of Date filter (a specific year, e.g. 2023, or All). The highest month was December at roughly \$79.2M, and the lowest was February at roughly \$23.8M.
- **Transaction Type:** a pie chart comparing Sale value (roughly \$479M) against Return value (roughly \$30.2M, shown as negative).
- **Revenue by Store:** a horizontal bar chart of the top 10 stores by revenue, led by Store Berlin at roughly \$25.1M.
- **Payment Method:** a pie chart comparing Cash revenue (roughly \$89.7M) against Credit Card revenue (roughly \$359M).

9.4 Product & Inventory Performance Dashboard

Focuses on product, category, and sub-category performance.

- Quantity by Category: a pie chart of quantity/revenue by main category — Feminine (roughly \$140.3M), Masculine (roughly \$118.6M), and Children (roughly \$41M).
- Revenue by Category: a bar chart comparing revenue by category — Feminine (roughly \$225.4M), Masculine (roughly \$175.6M), Children (roughly \$47.8M).
- Revenue by Sub Category: a horizontal bar chart ranked by revenue, led by Pants and Jeans (roughly \$49.8M), followed by Sportswear (roughly \$41.5M).
- Average Discount by Category: a column chart of average discount rate per category — Feminine 0.56%, Masculine 0.55%, Children 0.44%.

9.5 Customer & Geo-Insights Dashboard

Focuses on the geographic dimension and on customer and employee analysis.

- Revenue by Country: a filled map colored by revenue, topped by Germany (roughly \$68.7M), then Spain (roughly \$52.6M), then China (roughly \$8M).
- Revenue by City: a bar chart of the top 10 cities by revenue, led by Berlin (roughly \$18.7M).
- Top 10 Customers: a bar chart of the top 10 customers by total purchases (Invoice Total USD), led by a customer named Li Yang at roughly \$16,214.
- Revenue by Employee: a column chart of the top revenue-generating employees, led by Hannes Biggen at roughly \$3.18M.

9.6 Supporting Sheets

In addition to the sheets embedded directly in the three dashboards, a set of supporting sheets was built within the workbook: Total Revenue, Total Quantity, Total Discount, Total Stores, Sales Trend, Payment Method, Revenue by Sub Category, Revenue by City, Transaction Type, Revenue by Country, Quantity by Category, Average Discount by Category, Top 10 Customers, and Revenue by Employee. Each of these sheets is used as an object within one of the three dashboards described above.

Like the Power BI model, the Tableau workbook is built on a star schema with transactions.csv as the central fact table connected to five dimension tables via Relationships, giving decision-makers a second, complementary way to explore the same overall financial picture, product and inventory performance, and customer/geographic analysis.

10. Final Presentation & Reports

10.1 Deliverables Summary

Deliverable	Tool	Status
Data Cleaning & Modeling	Power BI / Power Query	Complete
DAX Calculated Columns & Measures	Power BI DAX	Complete
Power BI Dashboards (11 pages)	Power BI Desktop	Complete
Excel Tables, DAX & Stores Overview Dashboard	Microsoft Excel	Complete
Tableau Dashboard	Tableau	Complete
Project Documentation	Microsoft Word	Complete
GitHub Repository	GitHub	Complete

10.2 Key Findings

- Multi-currency transactions required USD standardization before any meaningful revenue comparison across regions — implemented consistently in both Power BI and Excel.
- Customer segmentation by Generation and Job Category enables targeted marketing analysis; Students and Gen Z represent the largest customer segments.
- Returning customers generate 69% of total revenue despite being a smaller share of the base, highlighting the value of retention.
- Store Tier C shows the highest sales volume but a comparatively lower margin, warranting operational review.
- Spain and China show the strongest year-over-year profit growth, while Portugal is declining.
- Discounted transactions represent 28.75% of volume but carry a noticeably lower profit margin (42.04%) than full-price sales (60.53%).

10.3 Recommendations

- Investigate the causes of Store Tier C's lower margin despite high sales volume.
- Use CLV segmentation to prioritize marketing spend on high-value customer generations.
- Leverage Job Category grouping to identify occupation-based purchasing trends.
- Review return rate drivers in Spain and the United Kingdom, the highest-return regions.
- Investigate the decline in Portugal's year-over-year profit growth.
- Consider adding a dedicated Date Dimension table to enhance time-intelligence DAX functions.